Progressive Attention-Based Feature Recovery With Scribble Supervision for Saliency Detection in Optical Remote Sensing Image

Xuan Li, Yuhang Xu, Lei Ma, Zhenghua Huang, and Haiwen Yuan

***Abstract*— Salient object detection (SOD) task for optical remote sensing images (RSIs) plays an important role in many remote sensing applications. Most of the existing methods train their networks depending on a large amount of pixel-wise datasets. However, such expensive and time-consuming training setting prevents the approaches becoming flexible and scalable solutions. To this end, we explore efficient SOD for optical RSIs based on easily accessible weak supervision source. In this work, we propose a novel end-to-end progressive attention-based feature recovery framework with scribble supervision. Specifically, to better locate challenging salient objects in optical RSIs, an object position module (OPM) is proposed to capture and enhance the long-range semantic dependence of objects' position information, which depends on the complementary attention mechanism. Also, to restore the entire salient objects, a context refinement module (CRM) is proposed, which extract local contextual information for better propagating high-level semantics to low-level details. Moreover, to improve the adaptability of the network to the changing scenarios of optical RSIs, we propose a salient region correcting (SRC) mechanism to help the predicted salient regions rectify their saliency values by constraining the saliency relationship between predictions from different augmentation models. In addition, due to the lack of dataset for weakly supervised SOD for optical RSIs, we relabeled an existing large-scale optical RSIs dataset with scribbles, namely, EORSSD-S. Experimental results on benchmark datasets demonstrate that the proposed method can outperform other weakly supervised SOD methods. Also, the proposed method even outperformed some fully supervised methods. https://github.com/melonless/PAFR.**

***Index Terms*— Convolutional neural networks (CNNs), optical remote sensing images (RSIs), salient object detection (SOD), weakly supervised.**

I. Introduction

With the significant development of space sensor and remote sensing technology, it has been easy to access massive amounts of optical remote sensing data. Due to the availability of such image data, objects analysis tasks in optical remote sensing images (RSIs) have recently gained a great attention [1], [2], [3], [4], [5], [6], which is needed in various military and civilian applications. Among these tasks, salient object detection (SOD) aims to capture the most attention-grabbing objects or regions in a given image [7], [8]. In the last decade, deep convolutional neural networks (CNNs) have become one of the most popular tools for SOD tasks. However, such methods often need large amount of pixel-wise annotations to work well. In the realm of remote sensing, optical RSIs tend to have largely different data distributions, since the various imaging conditions increase the data requirements for training stronger SOD models. Also, the task of labeling a large amount of pixel-wise salient objects in optical RSIs consumes too much time and labor cost. To overcome these limitations, it is essential to explore weakly supervised SOD methods for optical RSIs. It can dramatically reduce the cost of manual annotation and enable the models to be scale much easier than fully supervised methods [9].

Weakly supervised SOD methods based on image-level labels [10], [11] have achieved promising results in natural scene images (NSIs), which have simple background and definite objects. However, image-level labels cannot provide accurate location information for the network of SOD task; thus, it is hard to discriminate the most interesting objects form those easily confused objects and challenging scenes. In recent years, scribble annotation [12], [13], [14] has attracted much attention due to their flexibility and speed on labeling winding objects. As shown in Fig. 1, the scribbles on the foreground of image can directly provide the location information of objects, and the ones on the background can help the network ignore the unnecessary pixels of these regions.

Although scribble annotation is a useful weak supervision source to fast determine location information for the training of network, it is difficult to extract stable and effective salient objects, since most of the objects' structural and detailed information is missing. In the past few years, CNN-based methods [15], [16] introduced the "decoder" to obtain rich object information without bringing redundant supervision, which usually used low-level features from "encoder" to gradually restore the lost spatial information. For instance, convolutional layers were widespread used to integrate salient features from coarse to fine [17], [18], [19], [20]. However, such strategies often ignored the great differences of spatial

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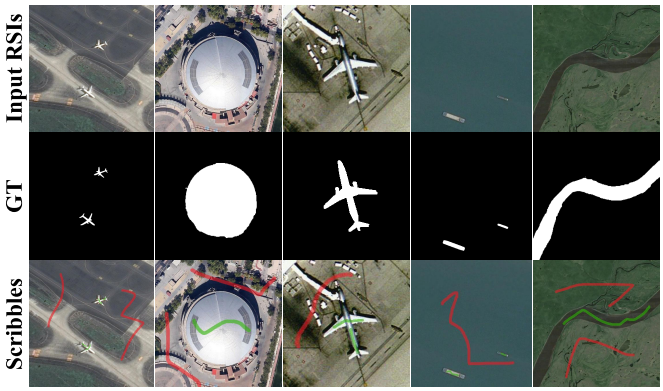


Fig. 1. Illustration of training data with different types of annotations. (Top to bottom) Input optical RSIs, dense mask annotations, and our scribble annotations. Scribbles colored in red and green represent foreground and background, respectively.

distribution and semantic information between low-level and high-level features. More importantly, different from NSIs photographed by manual adjustment, in the field of remote sensing, the challenge of SOD to optical RSIs usually stems from various imaging interferences (e.g., weather, illumination, and background patterns). Without explicit constraint on multilevel features, it is prone to depict amorphous salient objects in cluttered background with scribble weak supervision.

Thus, in this article, we propose a progressive attention-based feature recovery framework to achieve efficient SOD for optical RSIs with scribble weak supervision. Specifically, to extract complete salient objects from optical RSIs, we design an object position module (OPM) and a context refinement module (CRM) to boost information propagating for better feature recovery. The proposed OPM adopts a novel architecture design basing on spatial attention (SA) mechanism and channel attention (CA) mechanism. Elaborately, the SA block that captures SA is designed inside the CA block as a supplement to objects' position information. In this way, the OPM effectively simplifies the process of information transmission and makes full use of correlated position information of salient objects. Guided by the high-level attentive cues from OPM, the CRM is proposed to extract local contextual information for propagating high-level semantics to low-level details. With the multiscale network structure, it can restore complete salient objects in a progressive manner.

Moreover, to improve the adaptability of the network to the imaging interferences of optical RSIs, we propose a salient region correcting (SRC) mechanism for model training, which imposes explicit constraints on different saliency predictions. Since previous SOD methods use multiple data transformations in a single model to train their networks, they only impose limited constraint on network prediction. This prevents its applicability to large-scale and challenging optical RSIs. In our SRC mechanism, we simultaneously train two network branches (i.e., a native branch and an augmentation branch) in a collaboratively optimizing way. The augmentation branch contains multiple models, which share the same architecture and learn from training samples with different transformations, respectively. Furthermore, the native model can be updated by the augmentation models after each training batch.

To the best of our knowledge, there is limited number of works on training SOD models using weak supervision sources for optical RSIs. Due to the lack of scribble dataset for SOD tasks on optical RSIs, we use the existing largest and the most challenging dataset EORSSD [21] to relabel a new scribble dataset, called EORSSD-S. In summary, our main contributions are as follows.

1) An end-to-end progressive attention-based feature recovery framework is proposed to achieve weakly supervised SOD for optical RSIs, in which an OPM and a CRM are designed to boost the information propagating for better feature recovery.
2) We propose an SRC mechanism to improve the adaptability of the network to the imaging interferences of optical RSIs by constraining the saliency relationship between different predictions, for assisting the regions of restored side-output features to automatically calibrate their own saliency values.
3) We propose the first scribble-based optical RSIs dataset EORSSD-S to train our model. Also, the experimental results compared with 15 state-of-the-art methods demonstrate that the proposed method can achieve accurate salient objects localization and generate clear boundaries.

The rest of this article is organized as follows. In Section II, we review the recent development of SOD for NSIs, weakly supervised SOD methods for NSIs, and SOD methods for optical RSIs. In Section III, we present the proposed framework in detail. In Section IV, we conduct extensive experiments and ablation analyses. Finally, the conclusion is given in Section V.

II. Related Work

A. SOD for NSIs

In the last decade, traditional SOD methods have usually been driven by handcrafted local or global visual features, such as texture, color, and local/global contrast.

With the rapid development of deep learning, the performance of SOD models [22] has been significantly improved, and a large number of SOD models based on deep learning have been proposed. Wei *et al.* [23] designed a label decoupling framework (LDF) to decouple the original saliency map into body map and detail map. Liu *et al.* [24] proposed a stereoscopically attentive multiscale module to adaptively fuse the features of various scales using a stereoscopic attention mechanism. Chen *et al.* [25] developed a boundary-aware network for better detect salient objects using a contour loss and hierarchically global attention. However, these methods need to create large-scale pixel-wise labels, which are costly and time-consuming.

B. Weakly Supervised SOD for Optical RSIs

To reduce the labeling cost, weakly supervised SOD methods to NSIs have received much attention. Wang *et al.* [10] proposed a foreground inference network (FIN) and

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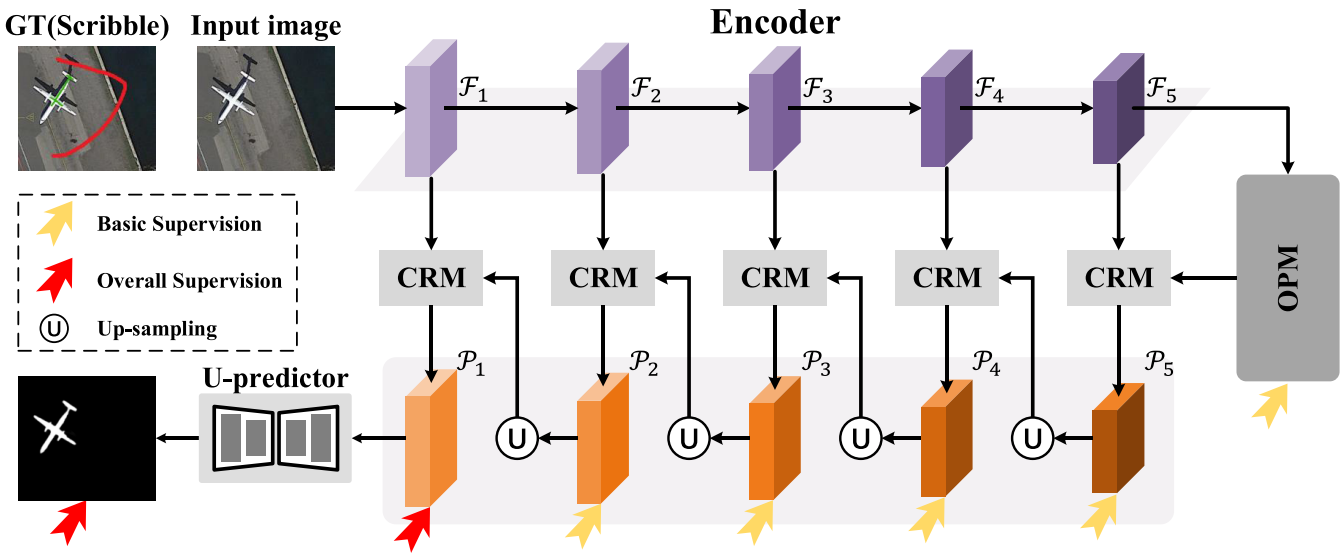


Fig. 2. Illustrate of our framework. The encoder extracts multilayer features $\mathcal{F}_i, i \in \{1, 2, 3, 4, 5\}$ from the input RSIs, which provide features for OPM and CRM. Specifically, OPM receives the top-level feature $\mathcal{F}_5$ as input and produces enhanced objects' location information to help the network focus on salient regions. CRMs take the current-level encoder features $\mathcal{F}_i$ to rectify the errors of $\mathcal{P}_{i+1}$. Finally, the generated saliency maps of network are supervised by the scribble annotations via the designed loss function.

trained it on image-level labels to produce saliency maps. Zhang *et al.* [12] explored SOD via scribble annotation and edge supervision source. In the realm of remote sensing, Zhang and Ma [26] designed a two-step framework to detect residential areas in RSIs. They used image-level labels to train a classification network to generate pseudolabels, which then were used to train the saliency network. Wei and Ji [27] introduced the dual-branch network to extract road surface in RSIs using both scribble and boundary supervision.

For saliency detection task in optical RSIs, image-level labels are too weak to detect complete salient objects, because they cannot provide sufficient objects' location information. Compared with image-level labels, scribble annotations can provide more definite location information of background and foreground, and just as important, the marking process of scribble is much simpler and more flexible. Thus, we utilize scribble annotation as an efficient weak supervision source.

C. Fully Supervised SOD for Optical RSIs

Compared with the SOD methods for NSIs, there is a few studies focus on SOD to optical RSIs. However, the SOD for optical RSIs is now a promising research trend. Zhang *et al.* [28] proposed a self-adaptively multiple feature fusion module using color, intensity, texture, and global contrast to produce saliency map for optical RSIs. In the method based on deep learning, Li *et al.* [29] designed a SOD method with an encoding–decoding structure based on the two-stream pyramid module and the nested connections. Li *et al.* [30] introduced a parallel down-up fusion network for SOD on optical RSIs using in-path and cross-path information. Zhang *et al.* [21] provided a dense attention fluid network for SOD in optical RSIs. They designed a dense attention fluid structure to integrate low-level cues into high-level attention maps and combined the global semantic information by a global context-aware attention module. Zhou *et al.* [31] designed an edge-guided network, which contains a shared encoder and two parallel decoders. They proposed an edge-aware position attention unit to take advantage of edge information and position attention features from the parallel decoders, so as to generate high-quality saliency maps. Although these methods based on deep learning have achieved amazing results, they all required a large number of manually labeled datasets and consumed too much labor and time cost.

III. Proposed Method

A. Overall Architecture

The overall framework of our proposed approach is shown in Fig. 2, which is an encoder–decoder architecture. The pipeline of the framework can be divided into three main stages: common feature extraction, objects position, and feature recovery. Given the input optical RSIs, we first use encoder to extract multilayer features. Then, a proposed OPM is implemented to receive the last layer output for extracting object position information in a global perspective. Finally, we propose CRM in each layer of the decoder to progressively restore the missing details. The output of the last CRM is finally passed to the U-predictor, which is a simple U-Net-like module flowing [22].

B. Feature Extractor

To detect size-various and type-various objects from complex background, the multiscale structure is necessary, since it can extract features with rich objects' information, which are helpful for depicting salient objects.

In detail, we use the designed encoder to extract multilayer features from the optical RSIs. The encoder contains five basic

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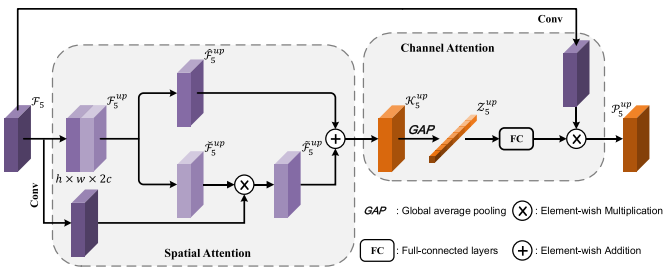


Fig. 3. Structure of the proposed OPM. The designed SA mechanism is applied to locate salient objects, and the embedded CA enhances the discrimination of objects position information.

stages, and the first four stages are composed of the first four layers of ResNet-34 [32]. Each stage has removed the pooling operation to generate feature maps with higher resolution in the earlier stage. To increase the overall receptive fields, we added a new stage after the fourth stage of the encoder, which is composed of three basic res-blocks and connected to the fourth stage through a max pool layer with a size of 2. For the input optical RSIs with the size of $H \times W$, we can get the $(H/16) \times (W/16)$ feature map from the encoder. Consequently, the ith side output $\mathcal{F}_i, i \in \{1, 2, 3, 4, 5\}$ from the encoder will be used as a guide for subsequent objects positioning and restoring of object details.

C. Object Position Module

In general, there are many interfering items in the background of optical RSIs. As shown in the first row of Fig. 1, the shadow, trees, and buildings in the background discard the accurately locating of cars. Some methods [33], [34], [35] consider used CA and SA mechanisms in a cascade way to enhance the expression capability of key semantic information, so as to locate potential targets. However, such strategies increase information redundancy and cause to inaccurate object location in some complex scenes. To make effective use of key semantic information, we propose an OPM. In OPM, we simplify the process of information transmission and design a simple but efficient complementary structure.

Fig. 3 shows the structure details of OPM. In the SA block, we use the interspatial relationship of input feature to enhance the ability of positioning important information, and then use the complementary CA block to boost channel-wise description of the semantic information. Specifically, given the lowest resolution feature as $\mathcal{F}_5 \in \mathbb{R}^{H \times W \times d}$, we first calculate the incomplete SA map $\mathcal{K}_5^{\text{up}}$ as follows:

$$\hat{\mathcal{F}}_5^{\text{up}}, \check{\mathcal{F}}_5^{\text{up}} = \text{conv}(\delta(\text{conv}(\mathcal{F}_5))) \tag{1}$$

$$\tilde{\mathcal{F}}_5^{\text{up}} = \text{conv}\left(\hat{\mathcal{F}}_5^{\text{up}}\right) * \text{conv}(\mathcal{F}_5) \tag{2}$$

$$\mathcal{K}_5^{\text{up}} = \delta\left(\check{\mathcal{F}}_5 + \tilde{\mathcal{F}}_5^{\text{up}}\right) \tag{3}$$

where conv is a 3×3 convolutional layer. δ denotes the ReLU activation function, and $*$ denotes the matrix multiplication.

Then, to enhance the representation ability of important information in $\mathcal{K}_5^{\text{up}}$, we first employ a global average pooling (GAP) on the incomplete SA map $\mathcal{K}_5^{\text{up}}$ to calculate the channel-wise global representation $\mathcal{Z}_5^{\text{up}} \in \mathbb{R}^{1 \times H \times W}$

$$\mathcal{Z}_5^{\text{up}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathcal{K}_5^{\text{up}}(i, j) \tag{4}$$

where H and W represent horizontal and vertical spatial size of $\mathcal{K}_5^{\text{up}}$, and (i, j) denotes the value at position (i, j) of $\mathcal{K}_5^{\text{up}}$. After that, we forward $\mathcal{Z}_5^{\text{up}}$ into a sample gated block to generate the spatial-channel weight $\Theta_{sc}$by learning nonlinear interactions and nonmuntual exclusive relationship between channels

$$\Theta_{\text{sc}} = \sigma\left(f_1\left(\delta\left(f_2\left(\mathcal{Z}_5^{\text{up}}\right)\right)\right)\right) \tag{5}$$

where $f_1 \in \mathbb{R}^{(C/r) \times C}$ and $f_2 \in \mathbb{R}^{C \times (C/r)}$ are two fully connected layers with the filter size of 1×1, and σ is the sigmoid function. Finally, we can get the global-aware feature maps $\mathcal{P}_5$ by applying channel-wise multiplication with $\mathcal{F}_5$ and Θ_{sc}

$$\mathcal{P}_5 = \Theta_{\text{sc}} * \mathcal{F}_5. \tag{6}$$

In this manner, we use this complementary structure to adaptively enhance the allocation of important information in the SA feature for guiding the positioning of complete objects more effectively.

D. Context Refinement Module

To restore complete salient object, the efficient interactions in multiscale and multilevel features are crucial. Because, the high-level features contain more abstract semantic information which are useful for locating complete salient regions, while the low-level features can provide more spatial details to restore spatial structure. Some methods [20], [22], [36], [37] progressively fuse the current-level features and the higher-level output using multiplication-summation. However, such strategies cannot achieve efficient information interactions [38] since there is more interference in the background of optic RSIs. In this work, we aim to conduct hierarchically context exploration for feature interaction, and then progressively restore complete object by fusing multiscale and multilevel features. For this purpose, we propose a novel CRM to capture local contextual information of hierarchical features through attention mechanism, so as to enhance the interactions of features. In this way, semantics and object details can be kept adaptively, which helps predict clearly challenging objects as shown in Fig. 4.

The details of CRM are shown in Fig. 5, each CRM takes the current-level feature from the encoder and the upsampled higher-level side-output as inputs, and produces refined features. Specifically, we first generate a coarse feature map $\mathcal{P}_i^1$ by applying element-wise multiplication to the side-out feature $\mathcal{P}_i$ and hierarchical feature $\mathcal{F}_i$, which can be written as follows:

$$\mathcal{P}_i^1 = \delta(\text{conv}(\mathcal{F}_i) * \text{conv}(\text{conv}(\mathcal{P}_i))) \tag{7}$$

where conv is a 3×3 convolutional layer. According to the importance of detailed information for detecting salient objects on optical RSIs, we attempt to leverage attention mechanism to perceive the context information for hierarchical

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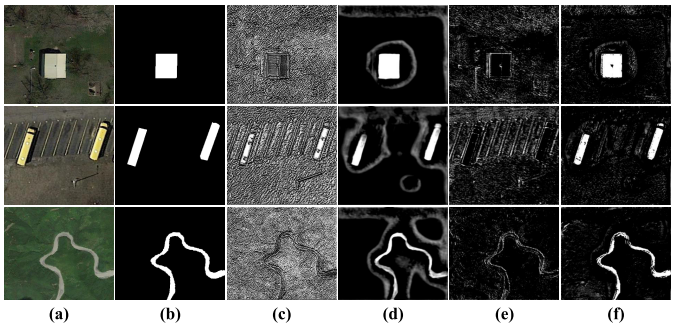


Fig. 4. Feature visualization in CRM. (a) Input RSIs. (b) Ground truth. (c) The input feature $\tilde{\mathcal{F}}_1$ from encoder. (d) Attentive feature $\tilde{\mathcal{F}}_1$. (e) Input feature $\tilde{\mathcal{P}}_1$ from previous CRM. (f) Output feature of CRM. We can see that CRM is very effective in restoring feature details.

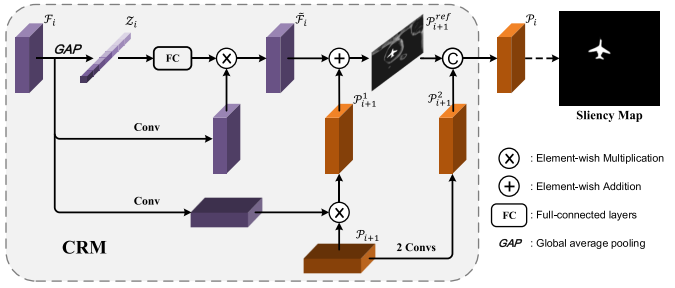


Fig. 5. Structure of the proposed CRM. The detailed information of low-level features are explicitly utilized for predicting fine-grained saliency maps.

features $\mathcal{F}_i (i = 1, 2, \ldots, 5)$. We perform a 3×3 convolution layer to compress the hierarchical features $\mathcal{F}_i$ to keep the same channel with the input feature from previous CRM, and then, we use the CA block to calculate the channel dependency inside the compressed $\mathcal{F}_i$. After that, we multiply it with $\mathcal{F}_i$ to obtain the attentive feature maps $\tilde{\mathcal{F}}_i$

$$\tilde{\mathcal{F}}_i = \mathrm{CA}(\mathrm{conv}(\mathcal{F}_i)) * \mathcal{F}_i \tag{8}$$

where $\mathrm{conv}(\cdot)$ is the combination of ReLU, batch normalization, and convolution layer. CA denotes the CA block, which uses the GAP to reduce the spatial error for getting a channel descriptor $\mathcal{Z}_i$, and then uses a gating block to obtain the channel weights.

Finally, we perform element-wise addition on the attentive feature maps $\tilde{\mathcal{F}}_i$ and $\mathcal{P}_i^1$ to restore the details. To maintain global semantic information, we concatenate it with the input feature $\mathcal{P}_i$ to generate restored global semantic feature $\mathcal{P}_{i-1}$

$$\mathcal{P}_i = \mathrm{concat}\left(\mathrm{conv}\left(\mathrm{conv}\left(\mathcal{P}_{i+1}^{\mathrm{ref}}\right)\right)\mathcal{P}_{i+1}\right) \tag{9}$$

where *concat* denotes the concatenate operation. In this way, the advantages of multilevel features can be effectively used to restore the details of salient objects.

E. SRC Mechanism

Through the designed framework, the complete salient objects are gradually restored, and the detection performance is greatly improved. However, different from NSIs photographed by manual adjustment, the challenge of SOD to optical RSIs usually stems from various imaging interferences

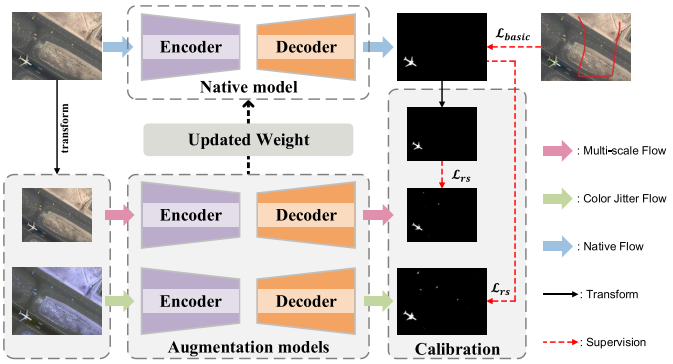


Fig. 6. Illustration of the SRC mechanism. During the test phase, we only use the native model to predict saliency maps for optical RSIs.

(e.g., weather, illumination, and background patterns) that change spectral characteristics of salient objects/regions. To improve the adaptability of the network to these interferences and depict more discriminative and complementary salient objects, we design an SRC mechanism using a reconstruction loss to constrain the saliency relationship between different predictions.

As shown in Fig. 6, in the training stage, the SRC mechanism first duplicates the model into two branches (i.e., a native branch and an augmentation branch). In the augmentation branch, multiple data transformations are applied on the input optical RSIs as interference conditions. With different transformed data, the models in augmentation branch can predict different saliency maps. Let S and $\tilde{S}$ denote the saliency prediction of the native model and the augmentation model, respectively. To constrain the saliency relationship between S and $\tilde{S}$ by regularize training. We then define the reconstruction loss between different saliency predictions as follows:

$$\mathcal{L}_{\mathrm{rs}} = \lambda_{\mathrm{ssim}}[1 - \mathrm{SSIM}(p, \tilde{p})] + \lambda_{\ell_1} \cdot \mathcal{L}^{\ell_1}\left(S_{i,j}, \tilde{S}_{i,j}\right) \tag{10}$$

where $S_{i,j}$ and $\tilde{S}_{i,j}$ denote the pixels from S and $\tilde{S}$, respectively. p is the center pixel of patch P from the saliency map S, so as to $\tilde{p}$. λ_{ssim} and λ_{ℓ_1} are weights to balance these two terms.

The SSIM term of (10) can be written as follows:

$$\mathrm{SSIM} = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \cdot \frac{2\sigma_{\mathrm{xy}} + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \tag{11}$$

where x and y are the pixel values of different patch, μ_x, μ_y and σ_x, σ_{xy} are their mean and standard deviations, and σ_{xy} is their covariance. $C_1 = 0.01^2$ and $C_2 = 0.03^2$ are used to avoid calculating the square root of zero.

When training through the SRC mechanism, the native model and the augmentation models are trained simultaneously. The augmentation model learns from the saliency prediction of native model and is optimized by $\mathcal{L}_{\mathrm{rs}}$, while the native model is optimized by $\mathcal{L}_{\mathrm{basic}}$. Besides, when training the network, the native model can be updated from the augmentation models that are optimized based on different data transformations. Furthermore, the reconstruction loss $\mathcal{L}_{\mathrm{rs}}$ imposes pixel-level and region-level constraints between the two branches, which benefits network training with regularization. As a result, it is expected that the network would

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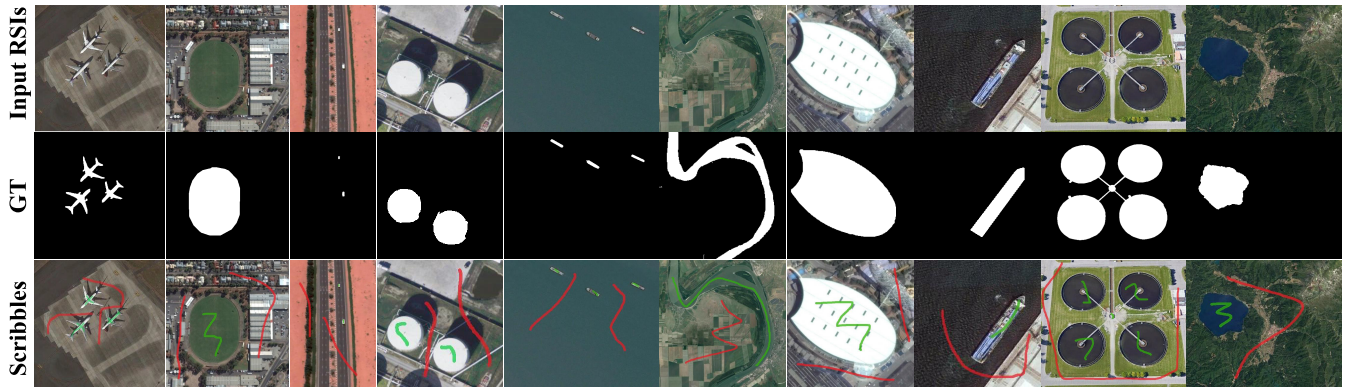


Fig. 7. Examples of our EORSSD-S dataset. (Top to bottom) Input optical RSIs, pixel-wise ground truth, and our scribble annotations. The green and red scribbles represent foreground and background, respectively.

perform better than a single native model trained only on scribble annotations. When testing our approach, we only use the native model to produce final saliency maps.

F. Loss Function

To make full use of the limited location information provided by scribble annotations, we introduce gated CRF loss [39] to provide more help at the boundary, denoted as L_{GCRF}. In other words, L_{GCRF} can effectively recover the object structure via scribble weak annotations, which can be written as follows:

$$\mathcal{L}_{\text{GCRF}} = \sum_{i} \sum_{j \in P} \psi_{\sigma_G}(i, j) \cdot K(i, j) \tag{12}$$

where P is a patch around pixel i, and $\psi_{\sigma_G}(\cdot)$ calculates the relationship between pixel i and j using a gate function. $K(\cdot)$ denotes the Gaussian kernel

$$K(i, j) = \omega \exp\left(-\frac{\| d_{(i)} - d_{(j)} \|^2}{2\sigma_\alpha^2} - \frac{\| I_{(i)} - I_{(j)} \|^2}{2\sigma_\beta^2}\right) \tag{13}$$

where $d_{(\cdot)}$ and $I_{(\cdot)}$ denote the position cues and color cues, respectively. ω, σ_α^2, and σ_β^2 are hyperparameters. Then, we use partial cross entropy loss to cooperate with $\mathcal{L}_{\text{rs}}$ and $\mathcal{L}_{\text{GCRF}}$, which can be written as follows:

$$\mathcal{L}_{\text{pce}} = \sum_{(x,y) \in \mathbb{S}} \mathcal{L}_{\text{ce}}(x, y) \tag{14}$$

where $\mathbb{S}$ represents the scribble dataset, and (x, y) is the position of pixel. $\mathcal{L}_{\text{ce}}$ denotes the cross entropy loss on (x, y). We use $\mathcal{L}_{\text{pce}}$ and $\mathcal{L}_{\text{GCRF}}$ to form the basic loss functions, which can be written as follows:

$$\mathcal{L}_{\text{basic}} = \mathcal{L}_{\text{pce}} + \beta \mathcal{L}_{\text{GCRF}} \tag{15}$$

where we set β to 0.3.

Finally, the overall loss is designed as follows:

$$\mathcal{L}_{\text{overall}} = \sum_{m=1}^{7} \mathcal{L}_{\text{basic}} + \sum_{m=6}^{7} \mathcal{L}_{\text{rs}} \tag{16}$$

where m denotes the mth outputs of the decoder.

IV. Experiments

A. Datasets

Since there is no scribble saliency dataset for SOD to optical RSIs, we relabeled an existing large-scale optical RSI SOD benchmark (i.e., EORSSD [21]) to train the proposed model. Different from fully supervised SOD methods trained with pixel-wise ground truth, we only use the scribble annotations as supervision. We refer the newly labeled dataset as EORSSD-S. Figs. 7 and 8 show some examples and statistics of our scribble annotations respectively, we simply annotate a few pixels of an image as "foreground" and "background." Then, we evaluate our method on two common optical RSIs datasets: ORSSD [29] and EORSSD [21]. ORSSD contains 800 challenging images, most of which are collected from Google Earth, of which 600 are used for training and 200 for testing. To enrich the scene and increase the challenge, EORSSD has expanded to 2000 pictures based on ORSSD, including 1400 for training and 600 for testing.

B. Saliency Evaluation Metrics

We adopt four evaluation metrics to comprehensively evaluate different methods: precision–recall (PR) curves, max F-measure score (F_β), S-measure score (S_m), and mean absolute error (MAE). The PR curves are calculated by comparing the binarized saliency maps with the ground truth. Also, the precision and recall are calculated from different binarization thresholds.

F-measure is the harmonic average of average precision and average recall, which aims to comprehensively measure precision and recall

$$F_\beta = \frac{\left(1 + \beta^2\right) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{17}$$

to emphasize the importance of precision, and β^2 is set to 0.3 [40].

MAE directly estimates the average absolute difference between prediction and corresponding ground truth on pixel-wise. The MAE is calculated as follows:

$$\text{MAE} = \frac{1}{H \times W} \sum_{u=1}^{W} \sum_{v=1}^{H} |P(u, v) - G(u, v)| \tag{18}$$

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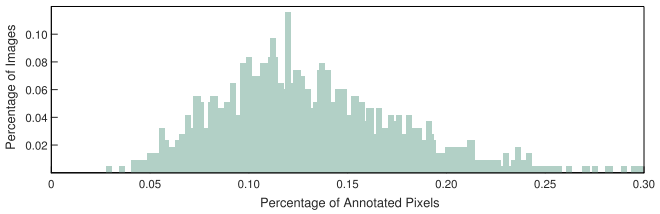


Fig. 8. Percentage of labeled pixels in our EORSSD-S dataset, including foreground pixels and background pixels.

where H and W denote the height and width of the input image, respectively. $P(u, v)$ and $G(u, v)$ are pixel coordinates from the predicted saliency map and its ground truth.

S-measure is used to compute the structural similarity between the predicted saliency map and its ground truth. The S-measure can be computed as follows:

$$S_m = \gamma \cdot S_o + (1 - \gamma) \cdot S_r \tag{19}$$

where we set γ to 0.5 [41]. S_o and S_r are the region-aware structural similarity and the object-aware structural similarity.

C. Implementation Details

We implement our network on Pytorch 1.8 and train it on a single NVIDIA GeForce RTX 3090 GPU. The loss weights λ_{ssim} and λ_{ℓ_1} in (10) are set to 0.85 and 0.15. The hyperparameters ω, σ_α^2, and σ_β^2 in (13) are set to 60, 5, and 3, respectively. In the training stage, the all images are resized to 256 $\times$ 256 and then randomly cropped to the size of 224 $\times$ 224. We train our network for 120 epochs using the Adam optimizer [42] with initial learning rate 1e-4, decay rate 0.9, decay epoch 80, and batch size 12. It takes about 8 h to train our approach. Different from some previous weakly supervised saliency detection methods, which were trained with additional supervision sources (e.g., edge map [12]) or complex model (e.g., pseudolabels [9], [43]), we only use scribble annotations for end-to-end training.

Note, the scribble annotations and the SRC mechanism are only used for model training. Thus, in the testing stage, we only apply our approach on the test images. We resize all the test images to the size of 256 $\times$ 256. The test images are directly fed into the network, generating saliency prediction with no additional postprocessing (e.g., CRF [10]).

D. Performance Comparison With State of the Art

To evaluate the model performance, we compare our proposed method with 15 state-of-the-art methods on two challenging datasets ORSSD [29] and EORSSD [21], including three unsupervised algorithms for NSIs: FES [44], GMR [45], and HDCT [46], four deep learning methods for NSIs: R3Net [47], PFAN [48], PoolNet [37], CPD [49], and EGNet [50], five deep learning methods for optical RSIs: VOS [51], CMC [52], LVNet [29], DAFNet [21], and RRNet [53], and two weakly supervised methods for NSIs: WSSA [12] and SCWS [13]. For fair comparison, we use the results directly provided by the authors or run their source code. For the fully supervised methods for NSIs, we retrain these methods on the training dataset of EORSSD under the same setting as the corresponding methods. For the weakly supervised methods for NSIs, we retrain these methods on our new proposed EORSSD-S dataset under the same setting as the corresponding methods.

1) Qualitative Evaluation: We compared our method with other state-of-the-art methods in Fig. 9, including fully supervised methods, which are specially designed for RSIs and other optical RSI methods or retraining methods using RSI datasets, and we showed the comparison results in some challenging scenarios (including aircraft, rivers, cars, buildings, and so on). It can be seen that compared with other methods, our method still has great advantages under the weak supervision setting, mainly including the following aspects.

a) Object positioning: Our method can effectively scan out where the challenging salient objects are and exclude the easily confused objects or regions. In disturbing scenes, such as the eighth and ninth rows, unsupervised and weakly supervised methods (e.g., GMR [45], WSSA [12], and SCWS [13]) can hardly eliminate the interference regions, and some fully supervised methods (e.g., CPD [49] and LVNet [29]) are not accurate enough in the description of salient objects. Relative to these methods, our method can clearly depict the complete salient region.

b) Object boundary: For the aircraft detection in the fourth row, we can see that our results have finer boundaries than the fully supervised methods (e.g., DAFNet [21], LVNet [29], and CPD [49]), and the weakly supervised methods (e.g., WSSA [12] and SCWS [13]) can only segment objects with rough boundaries. In the car detection in the sixth row, our results can completely depict the cars with clear boundaries, while other methods can only depict some parts of the car.

c) Challenge objects: For small objects in challenging scenes, our method is still highly sensitive to these objects with few detectable pixels. For example, in the eighth row, our method can predict the aircraft with high discrimination, while other methods cannot recover the details of the aircraft. In the seventh row, our method can accurately capture the salient object, while other methods only predict some fuzzy regions (e.g., DAFNet [21] and CPD [49]). For the winding rivers in the third and sixth rows, it is challenging to depict complete rivers, and some methods are hardly to capture the complete river or suppress the background regions, while the performance of our method is still competitive.

2) Quantitative Evaluation: We compared our method with 15 state-of-the-art methods on two optical RSI datasets and reported the complete quantitative comparison results in Table I. We can see that for unsupervised SOD methods, HDCT [46] achieved the best performance on MAE in both two datasets, but GMR [45] achieved the best performance in terms of S_m and F_β. We retrained some fully supervised SOD methods for NSIs on the training set of EORSSD, and we can see that their performance is much better than that of unsupervised SOD methods for NSIs; EGNet [50] achieves the best performance in terms of F_β, S_m, and MAE in these methods. For the fully supervised methods for optical RSIs, LVNet is a little worse than EGNet in terms of F_β

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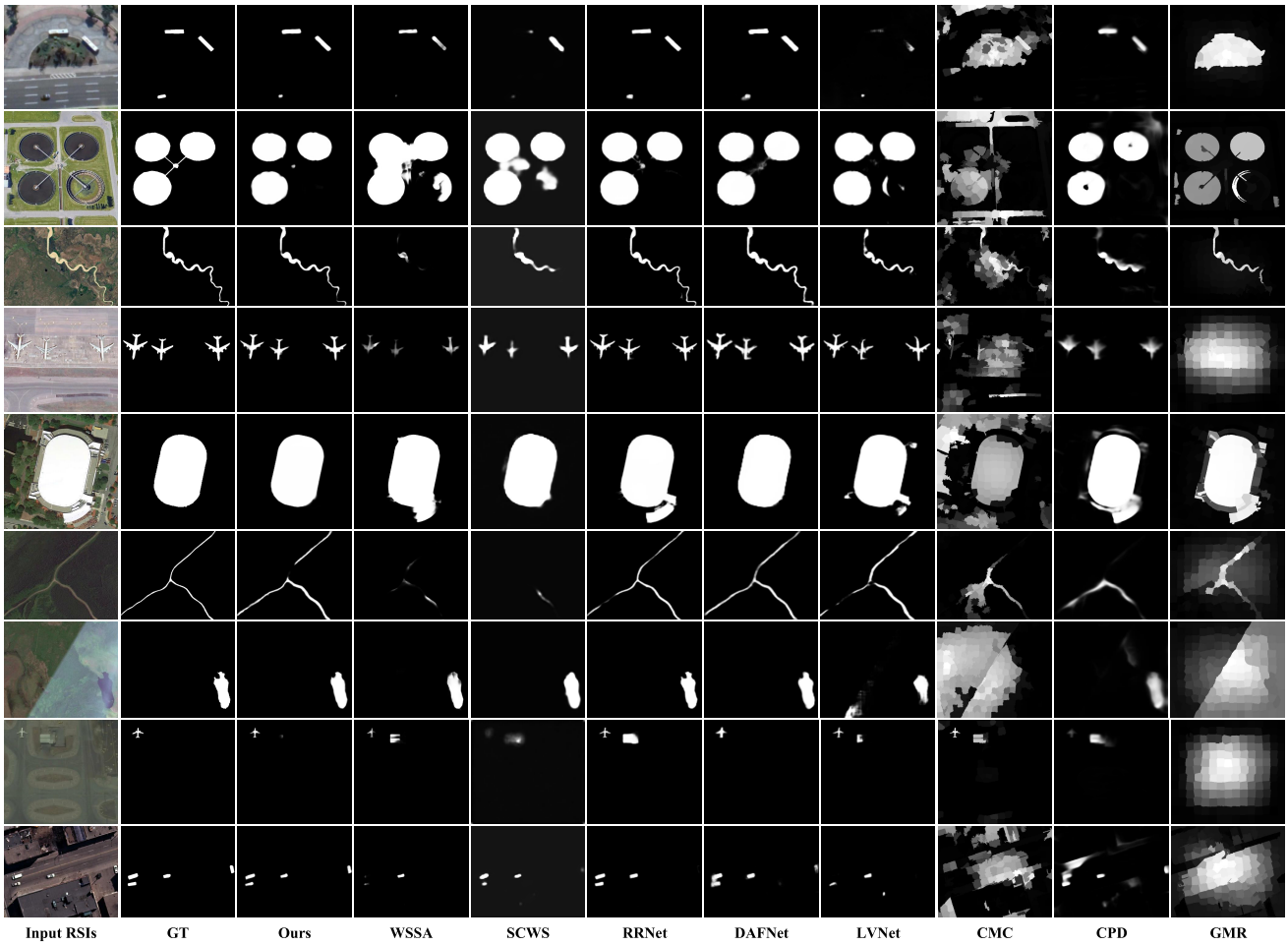


Fig. 9. Visual comparison of our method with other methods. The weakly supervised methods are retrained on the proposed EORSSD-S dataset, and the fully supervised methods are trained/retrained on the EORSSD [21] dataset.

TABLE I

QUANTITATIVE COMPARISON RESULTS WITH 15 STATE-OF-THE-ART METHODS (FULLY SUPERVISED AND WEAKLY SUPERVISED METHODS ARE MARKED WITH † AND ‡, RESPECTIVELY) ON THE TESTING SUBSET OF ORSSD [29] AND EORSSD [21] IN TERMS OF F_β, S_m, AND MAE. ↑ AND ↓ DENOTE THE LARGER AND SMALLER IS BETTER, RESPECTIVELY

Methods	ORSSD [29]			EORSSD [21]		
	F_β ↑	S_m ↑	MAE ↓	F_β ↑	S_m ↑	MAE ↓
FES [44]	0.455	0.585	0.097	0.308	0.567	0.065
GMR [45]	0.602	0.683	0.133	0.452	0.598	0.169
HDCT [46]	0.578	0.620	0.131	0.599	0.598	0.109
R3Net† [47]	0.782	0.814	0.040	0.771	0.819	0.017
PFAN† [48]	0.834	0.861	0.024	0.774	0.836	0.016
PoolNet† [37]	0.791	0.840	0.036	0.781	0.822	0.021
CPD† [49]	0.772	0.820	0.041	0.718	0.795	0.026
EGNet† [50]	0.844	0.872	0.022	0.806	0.860	0.011
VOS [51]	0.388	0.537	0.215	0.334	0.508	0.210
CMC [52]	0.421	0.603	0.127	0.366	0.580	0.106
LVNet† [29]	0.841	0.882	0.021	0.805	0.865	0.015
DAFNet† [21]	0.908	0.919	**0.011**	0.880	0.917	**0.005**
RRNet† [53]	**0.909**	**0.932**	0.012	**0.887**	**0.924**	0.008
WSSA‡ [12]	0.817	0.801	0.032	0.795	0.806	0.019
SCWS‡ [13]	0.767	0.701	0.097	0.700	0.639	0.089
Ours‡	**0.850**	**0.892**	**0.019**	**0.841**	**0.893**	**0.012**

and MAE, but perform better on S_m. RRNet [53] achieves the best performance on F_β and S_m, while DAFNet [21] is better on MAE. To verify the effectiveness of our method, we retrained two weakly supervised SOD methods for NSIs on the new proposed EORSSD-S (WSSA [12] and SCWS [13]), and we can see that our performance is significantly better than these methods on the three metrics in Table I and performs better than many fully supervised methods in segmentation of regions and boundaries.

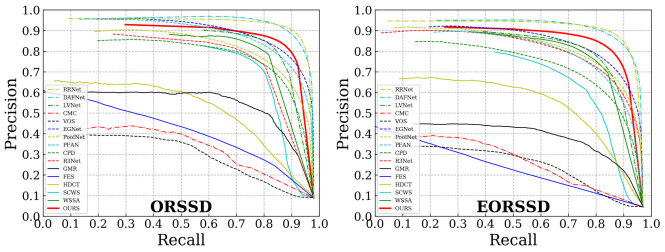


Fig. 10. PR curves of our method compared against 15 state-of-the-art SOD methods on ORSSD [29] and EORSSD [21]. The proposed method outperforms other weakly supervised methods and even some fully supervised methods.

In Fig. 10, we also plot the PR curves of our method compared with other methods on two optical RSI datasets to make a global comparison. We can see that the PR curves of our method are more closer to the fully supervised methods,

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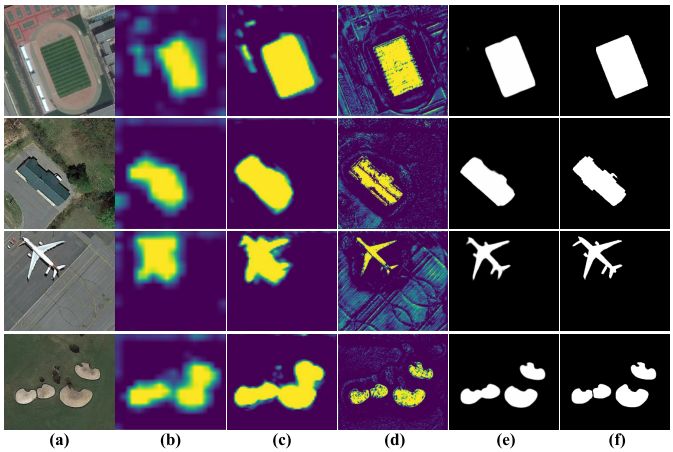


Fig. 11. Visualization of feature maps from the decoder. (a) Input RSIs. (b)–(d) Fifth, third, and first side outputs of the decoder. (e) Final predictions. (f) Ground truths.

TABLE II

ABLATION ANALYSIS FOR THE PROPOSED MODULE WITH DIFFERENT DISTRIBUTION CONFIGURATIONS ON ORSSD [29] AND EORSSD [21]. BASE IS THE ResNet-34 BACKBONE WITH A PREDICTION HEAD. OPM AND CRM ARE IMPLEMENTED IN THE DECODER TO IMPROVE THE NETWORK PREDICTION PERFORMANCE

No.	Module	ORSSD [29]		EORSSD [21]	
		$F_\beta \uparrow$	$S_m \uparrow$	$F_\beta \uparrow$	$S_m \uparrow$
1	Base	0.824	0.876	0.815	0.877
2	Base+OPM	0.836	0.881	0.821	0.881
3	Base+OPM+CRM	**0.841**	**0.887**	**0.829**	**0.889**
4	Finall	**0.850**	**0.892**	**0.841**	**0.893**

which means our methods perform better than most of the methods.

E. Ablation Study

To verify the effectiveness of different network configurations for our method, we perform comprehensive ablation study on two optical RSIs datasets, ORSSD [29] and EORSSD [21], including backbone network and loss function. We first analyze the contributions of OPM and CRM in the backbone and then compare the feature maps of different levels from the decoder with all components. Finally, we verify the effect of designed SRC mechanism.

1) Effectiveness of OPM and CRM: To verify the effectiveness of the proposed OPM and CRM, we compare the performance results under three configurations. We use the feature extractor as the baseline, which is denoted as "Base." To analyze the effectiveness of each component, we introduce OPM and CRM to configure the model.

The comparison results of different models are listed in Table II. As shown in the second row of Table II, by comparing the results of "Base," the addition of OPM obtains an average gain of 1.1% and 0.5% in terms of F_β and S_m on ORSSD [29] and EORSSD [21], respectively. Furthermore, to restore the object boundaries, we design CRM to configure the decoder. Besides, a better performance has been achieved by applying both OPM and CRM. After the addition of CRM, by comparing the results of "Base," the performance obtains an average gain of 1.9% and 1.3% in terms of F_β and S_m.

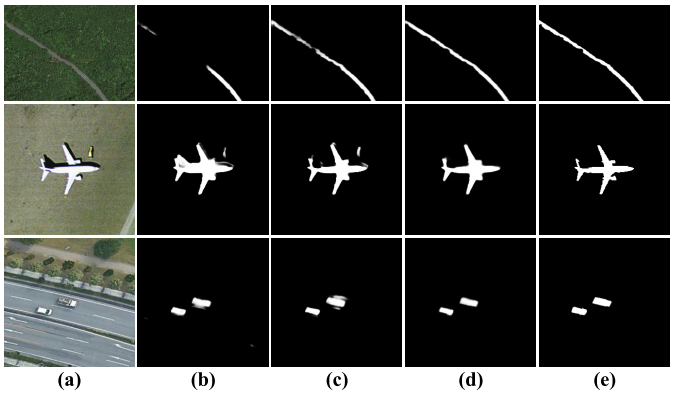


Fig. 12. Partial visualizations of our results with different components. (a) Input RSIs. (b)–(d) Results of baseline, baseline + OPM, and baseline + OPM + CRM. (e) Ground truth.

TABLE III

ABLATION ANALYSIS FOR THE DESIGNED SRC MECHANISM ON EORSSD [21]. THE BEST ONE IS COLORED IN RED

No.	Settings	$F_\beta \uparrow$	$S_m \uparrow$	MAE↓
1	None	0.814	0.876	0.021
2	scale change	0.833	0.883	0.014
3	color jitter	0.828	0.886	0.013
4	all (single model)	0.835	0.884	0.013
5	all (**ours**)	**0.841**	**0.893**	**0.012**

To intuitively show the optimization ability of the proposed OPM and CRM on prediction results, we show the prediction results under different configurations in Fig. 12. With the application of OPM, the network can depict salient objects and suppress the background (such as the river in the first row of Fig. 12) more accurately and effectively. By further adding CRM, the ability to describe detailed information has been greatly improved (such as the aircraft in the second row of Fig. 12).

2) Effectiveness of SRC Mechanism: To analyze the effect of the proposed SRC mechanism, we compare different training strategies. The first setting is to train the model with a single native model. To explore the respective effectiveness of different training strategies with data transformation (i.e., scale change and color jitter), we introduce different data transformation strategies for augmentation models.

Table III shows the comparison results of different training strategies by our SRC mechanism. When training with a single native model, the model can only learn from scribble annotations. Such a strategy obtains inferior results comparing with the results training with SRC mechanism. After introducing the SRC mechanism, the native model can be updated from the augmentation models that are optimized based on different data transformations. In addition, we can find that utilizing all the data transformations in a single augmentation model has a better performance improvement than only use one data transformation. Moreover, when all types of data transforma-

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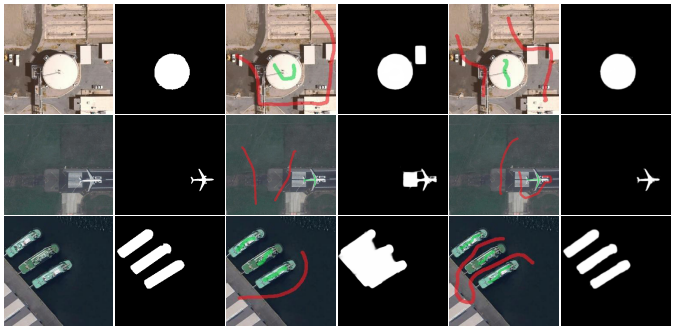


Fig. 13. Visualization of some typical ways of drawing scribbles and the generated saliency predictions. (Left to right) Input RSIs, pixel-wise ground truth, and initial and final scribble annotations with its generated predictions.

TABLE IV

ABLATION ANALYSIS FOR THE MODEL PERFORMANCE TRAINED WITH DIFFERENT SCRIBBLE ANNOTATIONS ON EORSSD [21]. THE BEST ONE IS COLORED IN RED

No.	Annotations	$F_\beta \uparrow$	$S_m \uparrow$	MAE↓
1	scribble 1 (initial)	0.826	0.882	0.014
2	scribble 2 (final)	**0.841**	**0.893**	**0.012**

tions are utilized in multiple augmentation models simultaneously, the performance of our SRC mechanism is further improved. Compared with other settings, our SRC achieves the best performance. Note, when testing our approach, we only use the native model to produce saliency maps for optical RSIs.

3) Effectiveness of Different Scribble Annotations: We analyze the model performance trained with different scribble annotations. In Fig. 13, we show some typical ways of drawing scribbles and the generated saliency predictions. When we first explore creating scribble annotations, scribbles are randomly drawn on the background and foreground of optical RSIs. But, it is proven that the scribble annotations influence the detection results. For instance, some regions in background are mistaken as salient objects for the ignorance of labeling, as shown in the first and second rows of Fig. 13. Besides, the network tends to predict neighboring regions as the same regions if they are not separated by scribbles, as shown in the third row of Fig. 13. So, we summarize the above failure ways of drawing scribbles and then recreate our scribble dataset. Table IV shows the model performance trained on different scribble annotations. For SOD in optical remote sensing, it can be seen that the quality of scribble annotations has a significant impact to the training performance.

The key criteria for creating high-quality scribbles are summarized as follows: 1) the nonsalient objects or regions, which are visually conspicuous, appearing in the background should be scribbled as background; 2) the imaging shadows and the interferences that attach to the salient objects should be scribbled as background; 3) when there are multiple objects, and they are close to each other, the background regions between them should be separated by scribbles; and 4) scribbles, such as simple straight lines or curves, labeled in the foreground should radially influence the scope of the whole salient objects.

TABLE V

QUANTITATIVE ANALYSIS ON THE OUTPUTS OF EACH LEVEL. $S_i, i \in (0, 1, 2, 3, 4, 5)$ MEANS THE iTH OUTPUT OF DECODER, AND S_a IS THE OUTPUT OF OPM. THE BEST ONE IS COLORED IN RED

Module	ORSSD [29]			EORSSD [21]		
	$F_\beta \uparrow$	$S_m \uparrow$	MAE↓	$F_\beta \uparrow$	$S_m \uparrow$	MAE↓
Sa	0.669	0.704	0.062	0.563	0.644	0.047
S5	0.683	0.729	0.052	0.559	0.669	0.040
S4	0.714	0.756	0.042	0.612	0.693	0.033
S3	0.803	0.811	0.031	0.740	0.759	0.022
S2	0.834	0.843	0.026	0.794	0.796	0.018
S1	0.850	0.890	0.020	0.832	0.881	0.013
S0	**0.850**	**0.892**	**0.019**	**0.841**	**0.893**	**0.012**

F. Model Analysis

Fig. 11 shows the visualization feature maps at different level of the decoder, we apply a 3×3 convolution layer and an up-sampling operation to the different level outputs to compress the feature channel and restore the feature size, and then produce the saliency map. To analyze the prediction performance of the proposed network, we conduct experiments on different levels of prediction results in terms of F_β, S_m, and MAE. As listed in Table V, the performance of prediction has been gradually improved under the supervision of network and gets the best performance from the zeroth output of decoder, which obtain an average gain of 38.22%, 30.59%, and 71.91% in terms of F_β, S_m, and MAE on ORSSD [29] and EORSSD [21] against the output of OPM.

V. CONCLUSION

The development of satellite remote sensing technology has enabled us to obtain a large number of RSI data. However, there are great differences between RSIs and NSIs in data distribution due to different imaging conditions. Also, the task of labeling commonly using pixel-wise annotations is usually time-consuming and laborious. It is essential to explore efficient SOD methods that can adapt to the growing requirement of processing remote sensing data.

In this article, we explored efficient SOD for optical RSIs based on progressive attention-based feature recovery with scribble weak supervision. We proposed an OPM to locate challenging objects using a complementary attention mechanism. Besides, we proposed a CRM to enhance the fusion of multiscale features by extracting local contextual information. Moreover, we designed an SRC mechanism to improve the adaptability of the network to the imaging interferences of optical RSIs via a reconstruction loss. Experimental results on two optical remote sensing datasets demonstrate that the proposed method can outperform other weakly supervised SOD methods. Furthermore, the proposed method even outperformed some fully supervised methods.

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